

Crows flexibly apply statistical inferences based on previous experience

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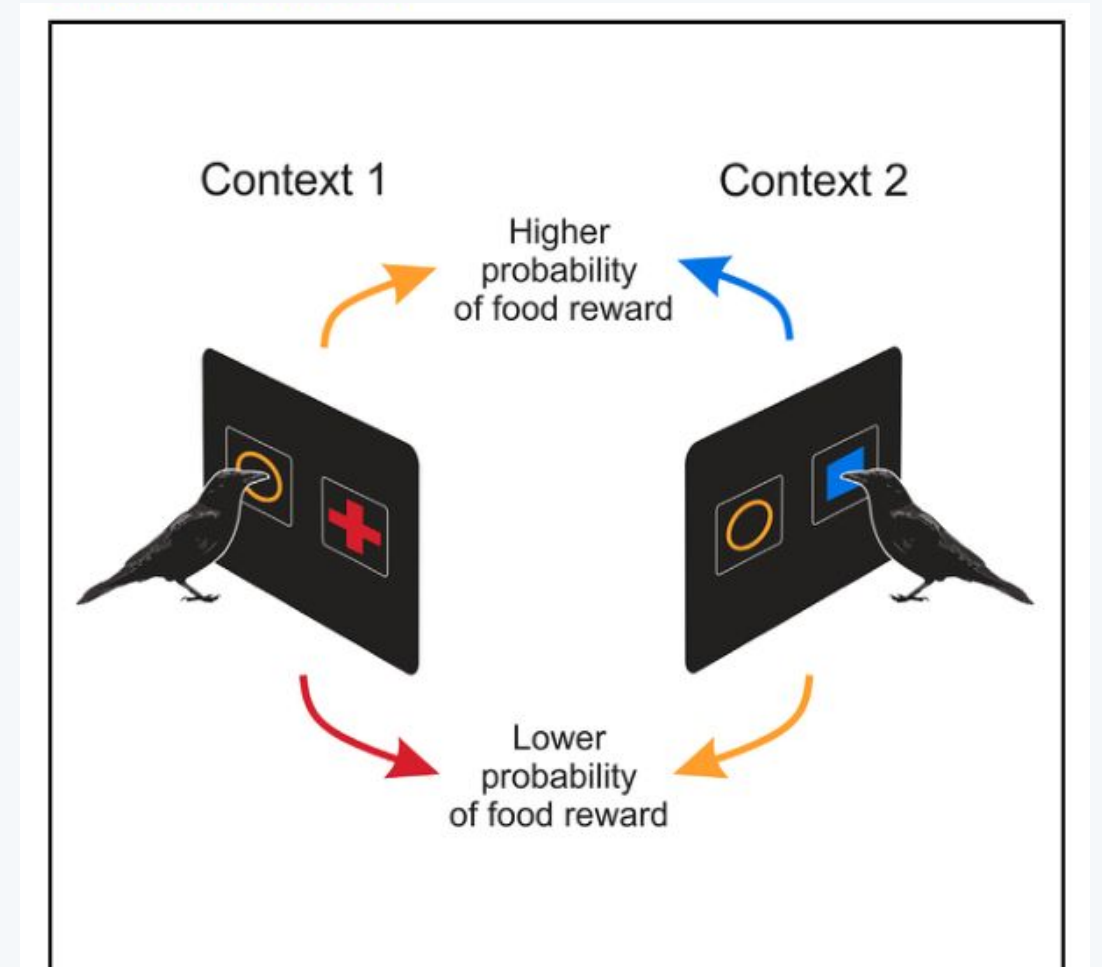
Outline

- Background
- Methods
- Results
- Discussion
- Critical Evaluation
- References

Background: what is statistical inference?

- The probability that an outcome will happen.
- Harder version of the task : remembering those chances from earlier experiences.

Why it matters :
Animals often forage under uncertainty.



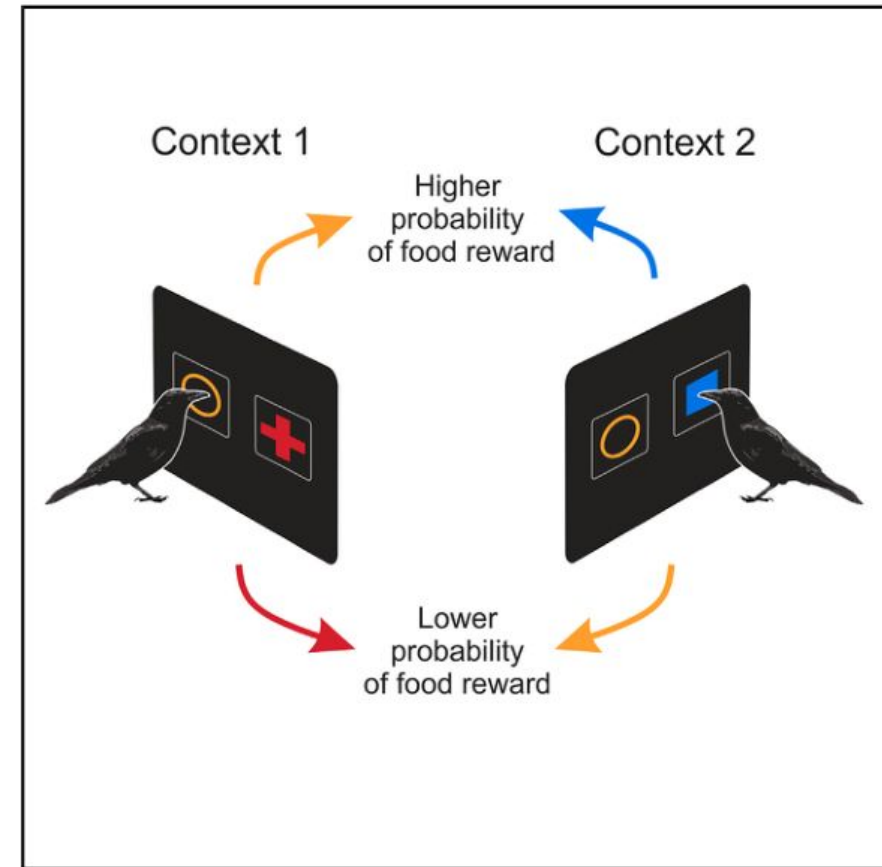
Why test crows?

- Carrion crows (*Corvus corone*) are corvids, a bird group known for flexible problem-solving and strong memory.
- Previous experience with touchscreen tasks.

Question:

Can crows remember several reward probabilities and flexibly choose the better option in new pairings?

Graphical abstract



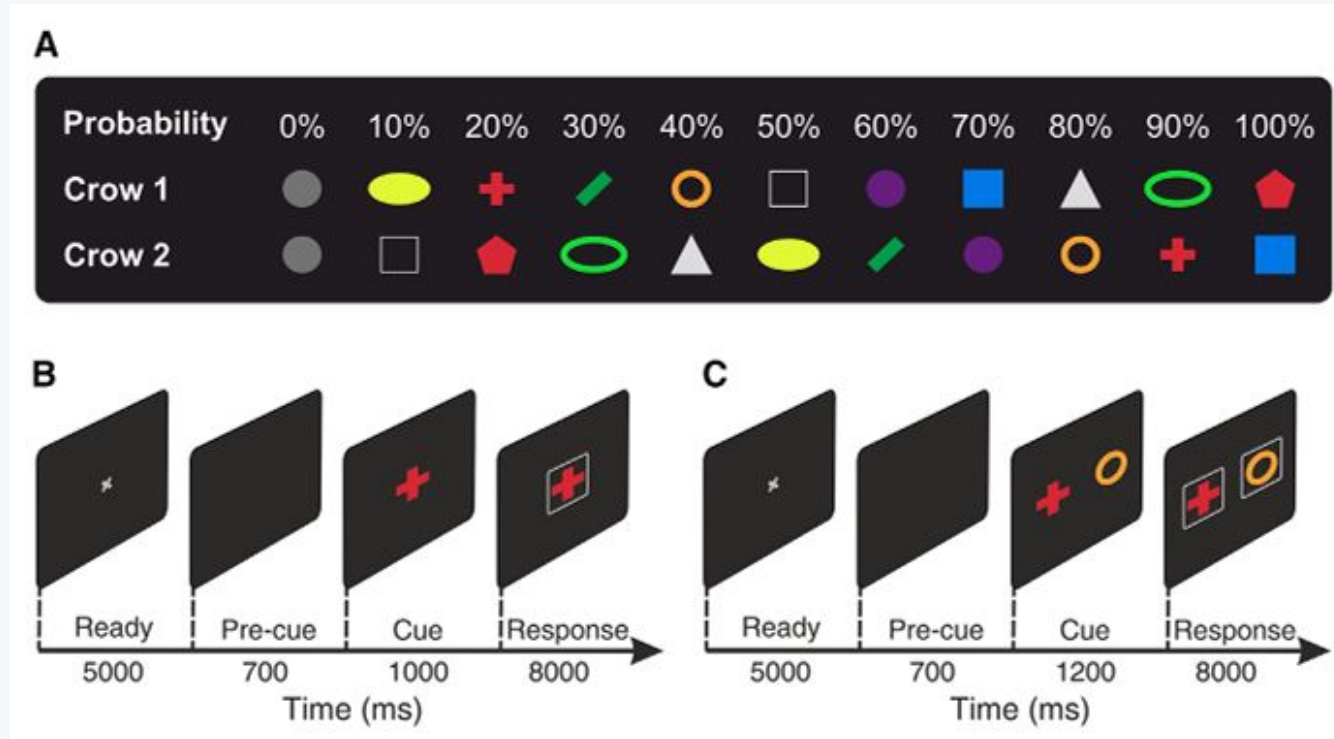
Methods: how the experiment worked

Step 1: learn the symbols

- Each crow learned 9 different symbols.
- Every symbol had a different chance of dispensing food.
- The symbols were arbitrary.

Step 2: forced-choice test

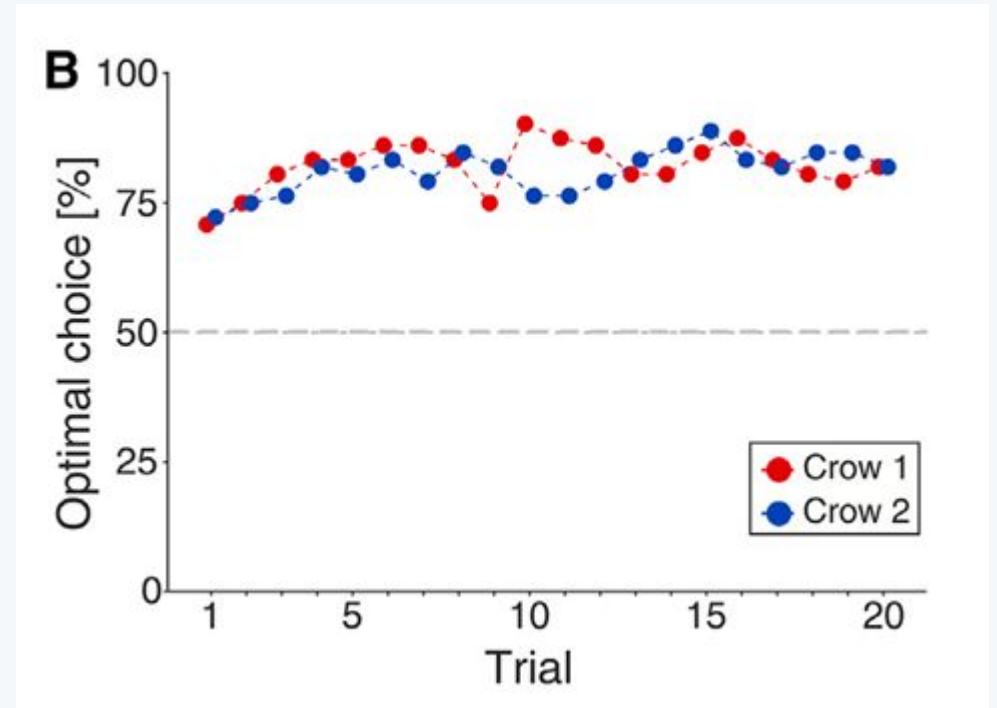
- Two symbols appeared together on the screen.
- The crow had to peck either symbol.



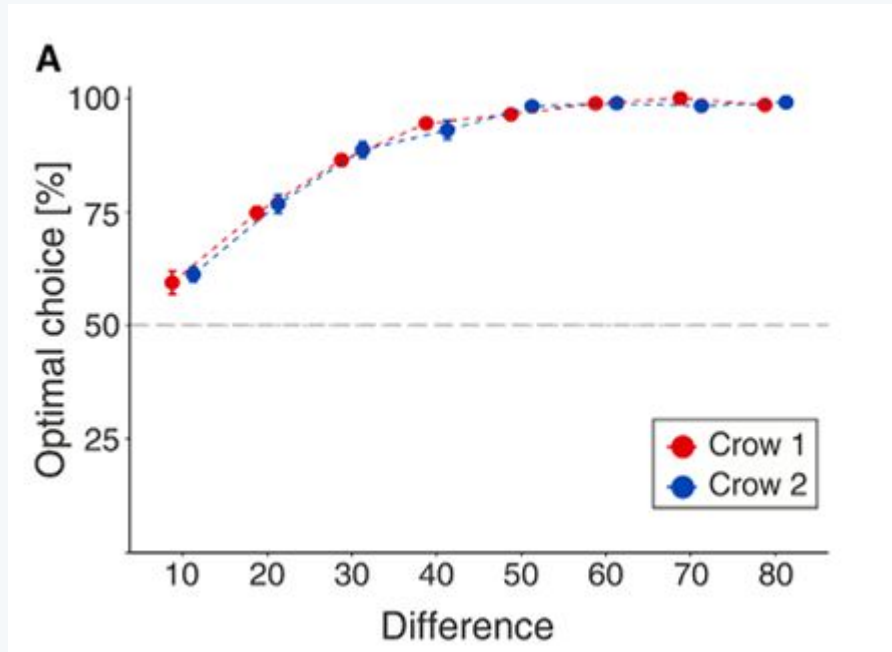
Results: the crows usually picked the better option

- Each crow was at or above 76% optimal choices across sessions.
- They also stayed accurate when retested about a month later.

In plain language: the birds learned which symbols were more reliable “bets”.



Results: the crows used probabilities flexibly



Two patterns were observed:

- Distance effect: difference between two probabilities
- Magnitude/ratio effect: the ratio between the two values also mattered.
- The crows seemed to treat probability as a mental quantity they could compare.

Why this matters:

The crows were not just memorizing “good” and “bad”. They compared probabilities in context.

Discussion: what does this mean in the real world?

What the study suggests

- Probabilities stored in memory.
- Comparing probabilities in new settings.
- Statistical reasoning under uncertainty.

Why that matters beyond crows

- Evidence of sophisticated reasoning
- Adaptive foraging decisions
- Probability linked to comparing quantities

Critical evaluation: what the study did well

1. Strong experimental control

- Arbitrary symbols
- Control task to test alternative explanation

2. Flexible reasoning was tested, not just simple memory

- 9 probabilities and many pairings
- Distance and ratio effects

Why these are effective design choices

- They improve internal validity
- They also make the conclusion more persuasive for a reader

Critical evaluation: limitations and open questions

1. Very small sample

2. The birds were unusual compared with wild crows

3. What the study cannot fully tell us

- The paper makes a strong case for flexible probability-based choice in these two trained crows.
- But the evidence is still best interpreted as promising and impressive rather than the final word.

References

- Johnston, M., Brecht, K. F., & Nieder, A. (2023). *Crows flexibly apply statistical inferences based on previous experience*. *Current Biology*, 33(15), 3238–3243. <https://doi.org/10.1016/j.cub.2023.06.023>